

Critical-order Schatten counterexamples from sparse phase space

Candidate counterexample packet for arXiv:2301.04044

12 August 2026

Status. Candidate counterexample, likely valid, awaiting specialist review. Open Problem 1.4 is false for every permitted parameter pair other than the classical endpoint $(\rho, \delta) = (1, 0)$. Open Problem 1.5 remains open.

1 Source problem

Cardona, Chatzakou, Ruzhansky, and Toft ask whether a non-elliptic operator

$$A \in \Psi_{\rho, \delta}^m(G), \quad 0 \leq \delta < \rho \leq 1,$$

that belongs to $\mathcal{S}_r(L^2(G))$ must have strict order $m < -n/r$, where $n = \dim G$ [1]. Their own exotic example at $(\rho, \delta) = (0, 0)$ is outside the strict parameter range of the problem.

$$\| \chi_N \circ A \chi_N \|_{L^1(G, \mathcal{S}_p(G))} = \int_G \| \chi_N \circ A \chi_N \|_{\mathcal{S}_p(G)}^{p(N)} \sim N^{n/p} \quad (1.1.1)$$

where $N > n$. Then $A \in \mathcal{S}_p(L^2(G))$ provided that $1 \leq p < \infty$.

1.4. Open problems. In view of the open questions that have arisen in our approach, and based on the active research on this field in the last 10 years, we propose the following open problems.

Open problem 1.4. Let $0 \leq \delta < \rho \leq 1$. Prove (or disprove) that if $A \in \Psi_{\rho, \delta}^m(G \times \widehat{G})$ is a non-elliptic pseudo-differential operator that belongs to the Schatten class $\mathcal{S}_r(L^2(G))$ then $m < -n/r$.

Open problem 1.5. Prove (or disprove) that for $r \in (1, \infty) \setminus \mathbb{Z}$, and with $A \in \mathcal{S}_r(L^2(G))$ being a non-elliptic operator, then the inequality $m \leq -n/r$ in (7) of Theorem 1.2 can be improved to the strict order estimate $m < -n/r$.

Remark 1.6. We observe that the equivalence $(1) \iff (3)$ in Theorem 1.1 was proved by the third author and J. Delgado in [28]. In particular, in [22] the relation between

Figure 1: Open Problems 1.4 and 1.5 on PDF page 6 of the source.

We use the standard toroidal quantization

$$\text{Op}(a)f(x) = \sum_{k \in \mathbb{Z}^n} e^{2\pi i k \cdot x} a(x, k) \widehat{f}(k).$$

The toroidal class $S_{\rho, \delta}^m$ is characterized by

$$|\partial_x^\beta \Delta_k^\alpha a(x, k)| \leq C_{\alpha, \beta} \langle k \rangle^{m - \rho|\alpha| + \delta|\beta|},$$

this agrees with the usual periodic Hörmander calculus [2].

2 Proof intuition

A symbol of order m need only reach size R^m somewhere at frequency scale R . When $\rho < 1$, the finite-difference estimates allow such a peak to occupy a ball of radius only R^ρ . Placing these balls at widely separated scales gives exact order while the number of active Fourier modes is only $R^{\rho n}$, not R^n . At the critical order $m = -n/r$, each packet then has Schatten mass $R^{-n(1-\rho)}$, which is summable.

At $\rho = 1$, a frequency packet has the full R^n modes. If $\delta > 0$, however, the symbol may be spatially concentrated on a set of volume $R^{-\delta n}$. The exact Hilbert–Schmidt formula sees this phase-space volume and again makes the critical-order series summable.

3 The counterexamples

Theorem 1. *Let $n \geq 1$.*

1. *If $r > 0$, $0 < \rho < 1$, and $0 \leq \delta < \rho$, there is a non-elliptic operator*

$$A \in \Psi_{\rho,\delta}^{-n/r}(\mathbb{T}^n) \setminus \bigcup_{\varepsilon>0} \Psi_{\rho,\delta}^{-n/r-\varepsilon}(\mathbb{T}^n)$$

such that $A \in \mathcal{S}_r(L^2(\mathbb{T}^n))$.

2. *If $0 < \delta < 1$, there is a non-elliptic operator*

$$A \in \Psi_{1,\delta}^{-n/2}(\mathbb{T}^n) \setminus \bigcup_{\varepsilon>0} \Psi_{1,\delta}^{-n/2-\varepsilon}(\mathbb{T}^n)$$

such that $A \in \mathcal{S}_2(L^2(\mathbb{T}^n))$.

Consequently Open Problem 1.4 is false for every allowed $(\rho, \delta) \neq (1, 0)$.

Proof. Fix a nonnegative $\chi \in C_c^\infty(\mathbb{R}^n)$, supported in a sufficiently small ball about zero, with $\chi(0) = 1$. Choose a large integer Q and put $R_j = Q^j$, starting at a sufficiently large index. All frequency packets below are then disjoint.

For the first assertion, put $m = -n/r$ and define a smooth function of the continuous frequency variable by

$$\tilde{a}(\xi) = \sum_j R_j^m \chi\left(\frac{\xi - R_j e_1}{R_j^\rho}\right), \quad \xi \in \mathbb{R}^n,$$

and let $a(k) = \tilde{a}(k)$ for $k \in \mathbb{Z}^n$. On the j -th packet, $\langle \xi \rangle \asymp R_j$ and

$$|\partial_\xi^\alpha \tilde{a}(\xi)| \leq C_\alpha R_j^{m-\rho|\alpha|} \lesssim \langle \xi \rangle^{m-\rho|\alpha|}.$$

Coordinate finite differences of a are iterated integrals of the corresponding derivatives of $\tilde{a}$. The same estimates therefore hold for $\Delta_k^\alpha a(k)$, including at packet boundaries. Since the symbol is independent of x , all positive-order spatial derivatives vanish. Thus $a \in S_{\rho,\delta}^m(\mathbb{T}^n \times \mathbb{Z}^n)$.

At the integer center $k_j = R_j e_1$,

$$a(k_j) = R_j^m.$$

If a had order $m - \varepsilon$, the zeroth symbol estimate at these centers would give $R_j^m \leq C R_j^{m-\varepsilon}$, or $R_j^\varepsilon \leq C$, a contradiction. Hence the displayed order is exact. The symbol also vanishes at arbitrarily large frequencies outside the packet union, so the operator is not elliptic.

The operator is a Fourier multiplier and is diagonal in the standard Fourier basis. Its singular values are the numbers $|a(k)|$. A packet contains at most $C R_j^{\rho n}$ lattice points, so

$$\|A\|_{\mathcal{S}_r}^r = \sum_{k \in \mathbb{Z}^n} |a(k)|^r \leq C \sum_j R_j^{mr+\rho n} = C \sum_j R_j^{-n(1-\rho)} < \infty.$$

This proves the first assertion.

For the second assertion, choose a coordinate chart κ about a point $x_0 \in \mathbb{T}^n$, with $\kappa(x_0) = 0$, and a function $\phi \in C_c^\infty(\mathbb{R}^n)$ supported strictly inside the chart, with $\phi(0) = 1$. Define ϕ_j on the torus by

$$\phi_j(x) = \phi(R_j^\delta \kappa(x))$$

inside the chart and zero outside it. After increasing the starting index, this is smooth and

$$|\partial_x^\beta \phi_j(x)| \leq C_\beta R_j^{\delta|\beta|}, \quad \int_{\mathbb{T}^n} |\phi_j(x)|^2 dx \leq C R_j^{-\delta n}.$$

With $m = -n/2$, set

$$a(x, k) = \sum_j R_j^m \phi_j(x) \chi\left(\frac{k - R_j e_1}{R_j}\right).$$

The same derivative-to-difference argument gives

$$|\partial_x^\beta \Delta_k^\alpha a(x, k)| \leq C_{\alpha, \beta} \langle k \rangle^{m - |\alpha| + \delta|\beta|}.$$

Thus $a \in S_{1, \delta}^m$. The identities $a(x_0, R_j e_1) = R_j^m$ prove exact order, and the large zero set proves non-ellipticity.

Finally, if $e_k(x) = e^{2\pi i k \cdot x}$, then $\text{Op}(a)e_k = e_k a(\cdot, k)$. Therefore

$$\|\text{Op}(a)\|_{\mathcal{S}_2}^2 = \sum_{k \in \mathbb{Z}^n} \|\text{Op}(a)e_k\|_2^2 = \int_{\mathbb{T}^n} \sum_{k \in \mathbb{Z}^n} |a(x, k)|^2 dx.$$

Disjointness of the frequency packets, their $O(R_j^n)$ lattice points, and the spatial-volume estimate give

$$\|\text{Op}(a)\|_{\mathcal{S}_2}^2 \leq C \sum_j R_j^{2m} R_j^n R_j^{-\delta n} = C \sum_j R_j^{-\delta n} < \infty.$$

This proves the second assertion and the stated consequence. $\square$

4 Verification, limitations, and novelty

The proof is exact and uses no numerical computation. The two decisive norm identities are elementary: diagonal multipliers have singular values $|a(k)|$, and the Hilbert–Schmidt norm is the squared L^2 norm of the symbol over $\mathbb{T}^n \times \mathbb{Z}^n$. The symbol-order checks use all finite differences and all spatial derivatives, not merely zeroth-order size.

The construction deliberately does not cover $(\rho, \delta) = (1, 0)$. At that endpoint, a critical frequency packet occupies R^n modes and no shrinking spatial volume is permitted. This is consistent with, but does not prove, the strict estimate in the source’s separate classical Open Problem 1.5.

A bounded primary-source search through 12 August 2026 used the source title and arXiv id, the exact strict-order phrase, non-elliptic Schatten pseudodifferential operators, and sparse toroidal Hörmander symbols. It found the source, the foundational toroidal quantization paper, and a later analogous Fourier-integral-operator problem, but no resolution or these constructions. Novelty confidence is moderate; specialist review is recommended before any priority claim.

References

- [1] D. Cardona, M. Chatzakou, M. Ruzhansky, and J. Toft, *Schatten-von Neumann properties for Hörmander classes on compact Lie groups*, arXiv:2301.04044 (2023).
- [2] M. Ruzhansky and V. Turunen, *Quantization of pseudo-differential operators on the torus*, arXiv:0805.2892 (2008).